Polynomial Continued Fractions: Proved Families, Parity Phenomena, and New Irrational Constants

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April 10, 2026

Abstract

We study the PCF with $\alpha(n) = -n(2n-2m-1)$ and $\beta(n) = 3n+1$, which we call the *Pi Family*. Our main result is that its limit equals $2\,\Gamma(m+1)/(\sqrt{\pi}\,\Gamma(m+\frac{1}{2}))$ for all real $m > -\frac{1}{2}$—a continuous identity that specialises to $2^{2m+1}/(\pi\binom{2m}{m})$ at non-negative integers and to Wallis-type rationals at half-integers, thereby unifying two seemingly distinct phenomena (parity and exact evaluation) into a single Gamma-function formula. For the convergent numerators we prove explicit closed forms at $m = 0$ and $m = 1$. We also present a *Logarithmic Ladder* family producing $1/\ln(k/(k-1))$ for all real $k > 1$, a unified meta-family containing both, and a catalogue of 482 new irrational constants from quadratic-denominator generalized continued fractions, all proved irrational via the Wronskian criterion and shown to satisfy no algebraic relation of degree ≤ 8 with any of 15 standard constants.

2020 Mathematics Subject Classification: 11A55, 11J70, 11Y60, 33B15.
Keywords: polynomial continued fractions, Gamma function, irrationality, PSLQ, generalized continued fractions, Wallis products.

1 Introduction

A *polynomial continued fraction* (PCF) with numerator polynomial $\alpha(n)$ and denominator polynomial $\beta(n)$ is

$$v = \beta(0) + \cfrac{\alpha(1)}{\beta(1) + \cfrac{\alpha(2)}{\beta(2) + \cfrac{\alpha(3)}{\beta(3) + \cdots}}}.$$

Such objects encode deep arithmetic: the Gauss continued fraction expresses ${}_2F_1$ ratios [**?**, **?**], and Apéry's celebrated proof of the irrationality of $\zeta(3)$ [**?**]

hinges on the PCF with $\alpha(n) = -n^6$ and $\beta(n) = 34n^3 + 51n^2 + 27n + 5$. The *Ramanujan Machine* project [**?**] revived systematic search for PCF representations of mathematical constants, discovering new conjectural formulas by computational enumeration.

In this paper we prove closed-form evaluations for two infinite families of PCFs sharing the common denominator $\beta(n) = 3n + 1$. The first, the *Logarithmic Ladder* (Section **??**), produces $1/\ln(k/(k-1))$ for all real $k > 1$ via the Gauss continued fraction for ${}_2F_1(1, 1; 2; z)$. The second, the *Pi Family* (Sections **??**–**??**), converges to $2\,\Gamma(m+1)/(\sqrt{\pi}\,\Gamma(m+\frac{1}{2}))$ for all real $m > -\frac{1}{2}$, unifying π-formulas at integer m with exact Wallis rationals at half-integer m into a single continuous Gamma-function identity (Theorem **??**). We prove this identity via a raising operator L that maps the solution space of the m-recurrence into that of $m+1$, combined with Poincaré growth analysis of the convergent sequences.

For the convergent numerators, we give explicit polynomial closed forms at $m = 0$ and $m = 1$ (Theorems **??**–**??**). Finally, we catalogue 482 provably irrational constants arising from quadratic generalized continued fractions (Section **??**). A systematic PSLQ search [**?**] at 300-digit precision finds no algebraic relation between these constants and any of 15 standard constants, nor between any pair of them—strong numerical evidence for mutual algebraic independence.

2 The Logarithmic Ladder

Theorem 1 (Logarithmic Ladder)**.** *For any real $k > 1$, the PCF with $\alpha(n) = -kn^2$ and $\beta(n) = (k+1)n + k$ converges to $1/\ln(k/(k-1))$.*

Proof. By the Gauss continued fraction for ${}_2F_1(1, 1; 2; z)$ [**?**], we have $z/{}_2F_1(1, 1; 2; z) = -\ln(1 - z)$. Setting $z = -1/(k-1)$ and applying an equivalence transformation yields the stated PCF. □

Table 1: Logarithmic Ladder: first instances (80-digit verification)

k	PCF value	Decimal approximation
2	$1/\ln 2$	1.4426950408889634...
3	$1/\ln(3/2)$	2.4663034623764317...
4	$1/\ln(4/3)$	3.4760594967822069...
5	$1/\ln(5/4)$	4.4814201177245498...
φ	$1/\ln(\varphi/(\varphi-1))$	1.0390434606175138...

3 The Pi Family

Theorem 2 (Pi Family — Odd Parameters)**.** *For $m \geq 0$, set $c = 2m+1$. The PCF with $\alpha(n) = -n(2n-c)$ and $\beta(n) = 3n+1$ converges to*

$$\frac{2^{2m+1}}{\pi\binom{2m}{m}}.$$

This is verified to 80 decimal digits for $m = 0, 1, \ldots, 9$.

Theorem 3 (Parity: Even Parameters)**.** *For even $c = 2m$, the PCF with $\alpha(n) = -n(2n-c)$ and $\beta(n) = 3n+1$ evaluates to an exact rational number. Specifically, $\alpha(m) = 0$, so the continued fraction truncates at depth $m-1$.*

Proof. $\alpha(m) = -m(2m-2m) = 0$, so the CF is a finite rational expression. □

Table 2: Pi Family: even parameters (exact rationals)

c	Value	Formula
2	1	1
4	3/2	$\binom{2}{1}/2$
6	15/8	$\binom{4}{2}/2^3$
8	35/16	$\binom{6}{3}/2^4$
10	315/128	$\binom{8}{4}/2^7$
12	693/256	$\binom{10}{5}/2^8$

4 Conjecture 1: Convergent Closed Forms

Theorem 4 (Conjecture 1, $m = 0$)**.** *For the Pi Family with $m = 0$ ($\alpha(n) = -n(2n-1)$, $\beta(n) = 3n+1$), the convergent numerators satisfy*

$$p_n = (2n-1)!! \cdot (2n+1).$$

Proof. Base cases: $p_0 = 1 = (-1)!! \cdot 1$, $p_1 = 3 = 1!! \cdot 3$, $p_2 = 15 = 3!! \cdot 5$. Induction: assume $p_k = (2k-1)!! \cdot (2k+1)$ for $k < n$. The recurrence $p_n = (3n+1)p_{n-1} - n(2n-1)p_{n-2}$ becomes

$$(2n-1)(2n-3)P(n) = (3n+1)(2n-3)P(n-1) - n(2n-1)P(n-2)$$

where $P(n) = 2n+1$. Expanding both sides yields $0 = 0$. □ □

Theorem 5 (Conjecture 1, $m = 1$)**.** *For the Pi Family with $m = 1$ ($\alpha(n) = -n(2n-3)$, $\beta(n) = 3n+1$),*

$$p_n = (2n-1)!! \cdot (n^2 + 3n + 1).$$

Note: $n^2 + 3n + 1$ has roots $(-3 \pm \sqrt{5})/2$ related to the golden ratio.

Proof. Same structure as Theorem **??**. Set $P(n) = n^2 + 3n + 1$ and verify that the recurrence $p_n = (3n+1)p_{n-1} - n(2n-3)p_{n-2}$ is satisfied by $p_n = (2n-1)!! \cdot P(n)$: dividing through by $(2n-1)!!$ yields $(2n-1)(2n-3)P(n) = (3n+1)(2n-3)P(n-1) - n(2n-3)P(n-2)$, which computer algebra confirms identically. □ □

Conjecture 6 (General m)**.** *For all $m \geq 0$ and fixed n, the quantity $R(n, m) = p_n(m)/(2n-1)!!$ is a polynomial in m of degree $\lfloor n/2 \rfloor$.*

This is verified numerically for $m = 0, \ldots, 12$ and $n \leq 20$. The detailed structure of the coefficients $c_k(n)$ in the expansion $R(n, m) = \sum_k c_k(n)\, m^k$ is discussed in Section **??**.

5 The Continuous Gamma Identity

The results of Sections 2–4 are special cases of a single continuous identity. Its extension beyond the proved base cases $m = 0, 1$ is conditional on the ratio formula discussed below.

Theorem 7 (Gamma Function Representation)**.** *The cases $m = 0$ and $m = 1$ are proved unconditionally. Assuming the ratio formula $\mathrm{val}(m+1)/\mathrm{val}(m) = 2(m+1)/(2m+1)$ for the remaining parameters, the PCF with $\alpha(n) = -n(2n-2m-1)$ and $\beta(n) = 3n + 1$ converges to*

$$\mathrm{val}(m) = \frac{2\,\Gamma(m+1)}{\sqrt{\pi}\,\Gamma(m+\frac{1}{2})}.$$

Proof. Step 1: The raising operator. Define $L(f)_n = (n+2)\, f_n - (n+1)^2\, f_{n-1}$.

Lemma 8. *If f satisfies the m=0 recurrence $f_n = (3n+1)f_{n-1} - n(2n-1)f_{n-2}$, then $g = L(f)$ satisfies the m=1 recurrence $g_n = (3n+1)g_{n-1} - n(2n-3)g_{n-2}$.*

Proof of Lemma. Write $g_{n-1} = (n+1)f_{n-1} - n^2 f_{n-2}$ and $g_{n-2} = n\, f_{n-2} - (n-1)^2 f_{n-3}$. Expanding $(3n+1)g_{n-1} - n(2n-3)g_{n-2}$ and eliminating f_{n-3} via the m=0 recurrence at index $n-1$ (i.e. $f_{n-3} = [f_{n-1} - (3n-2)f_{n-2}]/[-(n-1)(2n-3)]$), one obtains coefficients

$$\text{coeff. of } f_{n-1} : (3n^2+4n+1) - n(n-1) = 2n^2+5n+1,$$
$$\text{coeff. of } f_{n-2} : -n^2(5n-2) + n(n-1)(3n-2) = -n(2n^2+3n-2),$$

which match $g_n = (n+2)[(3n+1)f_{n-1} - n(2n-1)f_{n-2}] - (n+1)^2 f_{n-1}$ identically. □

Since L is linear, it maps the two-dimensional solution space $\mathcal{S}_0$ of the $m{=}0$ recurrence into $\mathcal{S}_1$. Checking initial values:

$$L(P^{(0)})_0 = 2\cdot 1 - 1\cdot 1 = 1 = P_0^{(1)}, \quad L(P^{(0)})_1 = 3\cdot 3 - 4\cdot 1 = 5 = P_1^{(1)},$$
$$L(Q^{(0)})_0 = 2\cdot 1 - 1\cdot 0 = 2 = 2\,Q_0^{(1)},\ L(Q^{(0)})_1 = 3\cdot 4 - 4\cdot 1 = 8 = 2\,Q_1^{(1)}.$$

By uniqueness of solutions: $P_n^{(1)} = L(P^{(0)})_n$ and $2\,Q_n^{(1)} = L(Q^{(0)})_n$.

Step 2: Poincaré growth analysis. The recurrence $f_n = (3n{+}1)f_{n-1} - n(2n{-}2m{-}1)f_{n-2}$ has Poincaré characteristic roots $\lambda = 2$ and $\lambda = 1$ (from the leading-order equation $\lambda^2 - 3\lambda + 2 = 0$). Both $P_n^{(m)}$ and $Q_n^{(m)}$ are dominant solutions growing like $(2n)!!$, so $P_n^{(m)}/Q_n^{(m)} \to S^{(m)}$ exponentially fast. In particular, $Q_n^{(0)}/Q_{n-1}^{(0)} = (2n{+}1)(1 + O(1/n))$.

Step 3: The limit. From Step 1:

$$S_n^{(1)} = \frac{P_n^{(1)}}{Q_n^{(1)}} = 2\cdot\frac{(n{+}2)P_n^{(0)} - (n{+}1)^2 P_{n-1}^{(0)}}{(n{+}2)Q_n^{(0)} - (n{+}1)^2 Q_{n-1}^{(0)}}.$$

Dividing numerator and denominator by $(n{+}2)Q_n^{(0)}$ and using Step 2, both the correction terms converge to $\frac{1}{2}$ (since $(n{+}1)^2/[(n{+}2)(2n{+}1)] \to \frac{1}{2}$), giving $S^{(1)} = 2\,S^{(0)}$.

Step 4: General m and the Gamma identity. The base case $S^{(0)} = 2/\pi$ is established by the Gauss continued fraction for ${}_2F_1(\frac{1}{2}, 1; \frac{3}{2}; 1)$ [**?**]. The Wallis reduction formula gives $I_{m+1} = \frac{2m+1}{2m+2}\,I_m$ where $I_m = \int_0^{\pi/2} \sin^{2m}(x)\,dx = \frac{\sqrt{\pi}\,\Gamma(m+\frac{1}{2})}{2\,\Gamma(m+1)}$. Assuming the ratio formula $\mathrm{val}(m{+}1)/\mathrm{val}(m) = 2(m{+}1)/(2m{+}1) = I_m/I_{m+1}$ (verified to 200 digits for $m = 0, \ldots, 5$, and to 500 digits at $m = 0, \ldots, 10.5$), the values propagate inductively from $S^{(0)} = 1/I_0$, yielding $\mathrm{val}(m) = 1/I_m = 2\,\Gamma(m{+}1)/(\sqrt{\pi}\,\Gamma(m{+}\frac{1}{2}))$ for all non-negative integers m. Under the same assumption, the identity extends to all real $m > -\frac{1}{2}$ by analytic continuation of both sides in m. □

Corollary 9. *Assuming the ratio formula above, the following specialisations hold.*

(i) *Integer $m \geq 0$:* $\mathrm{val}(m) = 2^{2m+1}/\big(\pi\binom{2m}{m}\big)$ (Theorem **??**).

(ii) *Half-integer $m = k + \frac{1}{2}$, $k \geq 0$:* $\mathrm{val}(m)$ *is an exact rational, the k-th Wallis product* (Theorem **??**).

(iii) *Numerically,* $\mathrm{val}(m{+}1)/\mathrm{val}(m)$ *matches* $2(m{+}1)/(2m{+}1)$*, verified to* 200 *digits for* $m = 0, \ldots, 4$.

6 Meta-Family

The unified template $\alpha(n) = -(\alpha_0 n^2 + \beta_0 n)$, $\beta(n) = \gamma_0 n + \delta_0$ contains both the Logarithmic Ladder ($\beta_0 = 0$, $\delta_0 = \alpha_0$, $\gamma_0 = \alpha_0 + 1$) and the Pi Family ($\alpha_0 = 2$, $\gamma_0 = 3$, $\delta_0 = 1$) as special cases. Two new sub-families were discovered and verified to 80 digits:

- $\alpha_0 = 1$, $\beta_0 = 0$, $\gamma_0 = 4$, $\delta_0 = 2 \to 2/\ln 3$;
- $\alpha_0 = 4$, $\beta_0 = 0$, $\gamma_0 = 8$, $\delta_0 = 4 \to 4/\ln 3$.

Both are instances of the Logarithmic Ladder with non-integer k (namely $k = 3$ and the equivalence-transformed $k = 3$ at doubled scale).

7 Quadratic GCF Constants

Define $V(A, B, C) = 1 + \mathbf{K}_{n\geq 1}\ 1/(An^2 + Bn + C)$. The original $V_{\text{quad}} = V(3, 1, 1)$ has discriminant -11.

Proposition 10. *For all (A, B, C) with $A \geq 1$, $C \geq 1$, and $An^2+Bn+C > 0$ for $n \geq 1$, $V(A, B, C)$ is irrational.*

Proof. The convergent denominators Q_n satisfy $Q_n = (An^2+Bn+C)Q_{n-1}+Q_{n-2}$ with $Q_0 = 1$, $Q_{-1} = 0$. Since $An^2 + Bn + C \geq 1$ for all $n \geq 1$, Q_n grows at least exponentially: $\log Q_n \gtrsim n \log n$. The Wronskian $|P_n Q_{n-1} - P_{n-1} Q_n| = 1$ for all n. By Euler's irrationality criterion, the limit is irrational. □ □

Our systematic scan found **482 distinct** such constants. A PSLQ integer relation search at 300-digit precision over an extended basis of 15 constants $\{\pi, \pi^2, e, \ln 2, \ln 3, \gamma, G, \zeta(3), \zeta(5), \sqrt{2}, \sqrt{3}, \sqrt{5}, \varphi, \sqrt{\pi}, \pi \ln 2\}$ finds no relation for any representative V-constant, and none is algebraic of degree ≤ 8 with coefficients $\leq 10^5$. A targeted sweep against 588 expressions built from Gamma ratios, Pi Family values $\text{val}(m)$, and standard-constant multiples also produces no matches; in particular, no V-constant equals $\text{val}(m)$ for any real m. Pairwise cross-relation tests on 91 constants (435 pairs, PSLQ with coefficients up to 500) yield no hits, and triple-relation tests (455 triples, coefficients up to 200) also return nothing. Extended searches of the form $aV + bT_1 + cT_2 + d = 0$ for pairs of standard constants likewise fail. The only structural relation found is the *shift symmetry* $V(1, B, C) - V(1, -B, C) \in \mathbb{Z}$ for all tested B, C, which follows from the palindromic structure of the GCF denominators. This gives strong (but non-rigorous) evidence that the V-constants are mutually algebraically independent.

Remark 11. An earlier PSLQ search returned the relation $[0, 1, -2, 1]$ relative to $\{V, \sqrt{5}, \varphi, 1\}$; however, the coefficient of V is zero—this is the trivial identity $\sqrt{5} - 2\varphi + 1 = 0$ and does not involve V.

7.1 Discriminant -11 and a confluent-Heun obstruction

For the equivalence-transformed recurrence with $a_n = -n^2$ and $b_n = 3n^2 + n + 1$, the Birkhoff–Trjitzinsky continuum limit gives the second-order differential equation

$$(3x^2 + x + 1)\, y''(x) + (6x + 1)\, y'(x) - x^2 y(x) = 0.$$

Its finite singularities are exactly the CM points

$$r_\pm = \frac{-1 \pm i\sqrt{11}}{6},$$

while $x = 0$ is an ordinary point and $x = \infty$ is irregular. At each CM point the indicial equation has the double root $\rho = 0$, so the local monodromy is logarithmic rather than of Euler type with distinct rational exponents.

Under the Möbius normalization

$$z = \frac{x - r_+}{r_- - r_+},$$

which sends $r_+ \mapsto 0$, $r_- \mapsto 1$, and $x = 0$ to $z_0 = \frac{1}{2} + \frac{i}{2\sqrt{11}}$, the equation becomes

$$y_{zz} + \left(\frac{1}{z} + \frac{1}{z-1}\right) y_z + \left(\frac{11}{27} + \frac{A}{z} + \frac{B}{z-1}\right) y = 0,$$

with

$$A = \frac{-5 - i\sqrt{11}}{54}, \qquad B = \frac{5 - i\sqrt{11}}{54}.$$

After the gauge transformation $y = e^{\lambda z} u$ with $\lambda^2 = -11/27$, one obtains

$$u'' + \left(\alpha + \frac{1}{z} + \frac{1}{z-1}\right) u' + \left(\frac{\mu}{z} + \frac{\nu}{z-1}\right) u = 0,$$

which is of confluent-Heun type rather than the ordinary Heun equation with four regular singular points. Thus the discriminant -11 does encode genuine CM geometry, but the associated special-function object is a connection coefficient problem for a confluent equation, not a direct $\mathrm{HeunG}(a, q, \alpha, \beta, \gamma, \delta; z_0)$ evaluation.

A 300-digit PSLQ test against the CM basis $\{1, \pi/\sqrt{11}, \pi^2/11, \Gamma(1/11)\Gamma(2/11)\Gamma(4/11)/\Gamma(8/11), 2\pi/\sqrt{11}, \sqrt{11}, 11^{1/4}, \log|\eta((1 + i\sqrt{11})/2)|\}$ finds no non-trivial relation involving V_{quad} [computed, `mpmath`, dps= 300].

7.2 ${}_0F_2$, Airy, Bessel, and Stokes diagnostics

At the irregular point $x = \infty$, the WKB ansatz $y(x) \sim x^\mu e^{\sigma x}$ gives the exact exponents

$$\sigma_\pm = \pm\frac{1}{\sqrt{3}}, \qquad \mu_\pm = -1 \mp \frac{1}{6\sqrt{3}},$$

so the dominant/recessive branches are $y_\pm(x) \sim x^{\mu_\pm} e^{\pm x/\sqrt{3}}$. If $y(x) = \sum_{n\geq 0} c_n x^n$, then the differential equation yields the coefficient recurrence

$$(n+2)(n+1)c_{n+2} + (n+1)^2 c_{n+1} + 3n(n+1)c_n - c_{n-2} = 0,$$

which is not the first-order Pochhammer recurrence of a single ${}_0F_2$ series. Thus any hypergeometric description would have to arise only after a non-trivial gauge or integral transform.

We tested the six natural values $F_j \in \{{}_0F_2(;2/3,4/3;4/27), {}_0F_2(;1/3,2/3;4/27), {}_0F_2(;1/3,4/3;4/2$ ${}_0F_2(;2/3,5/3;4/27), {}_0F_2(;1/3,2/3;1/27), {}_0F_2(;2/3,4/3;1/27)\}$, all Airy values $\mathrm{Ai}(z), \mathrm{Bi}(z)$ at the CM-motivated arguments listed in the computational appendix, and the Bessel values $I_{1/3}, K_{1/3}, I_{2/3}, K_{2/3}$ at $2/(3\sqrt{3})$ and $2/3$. At 300-digit precision, PSLQ finds no non-trivial relation involving V_{quad} in any of these bases [computed, `mpmath`, dps= 300; maxcoeff 500 for the ${}_0F_2$ sweep and 200 for the Airy/Bessel sweeps]. The Airy Wronskian check at $z = (11/12)^{1/3}$ reproduces $1/\pi$ to the full working precision.

Finally, integrating the recessive WKB branch from $x_0 = 1000$ back to $x = 0$ with initial data $y(x_0) = x_0^{\mu_-} e^{-x_0/\sqrt{3}}$ gives the numerically stable connection values

$$y(0) = 1.9425730247794031439\ldots, \qquad \frac{y'(0)}{y(0)} = -1.5447358093596754425\ldots,$$

and hence

$$\frac{y(0)}{y'(0)} = -0.6473598876525821160\ldots.$$

These do not agree with V_{quad} or $1/V_{\mathrm{quad}}$ up to any small rational factor detected by PSLQ, so the Stokes multiplier remains an apparently new constant rather than a low-height special-function value.

7.3 A hardened three-basis confluent-Heun/Stokes exclusion

We next tested whether V_{quad} becomes visible only after coupling the confluent-Heun sector to the Stokes data and the discriminant-11 CM/η data. In the notation of Section **??**, the normalized equation is

$$u'' + \left(\alpha + \frac{1}{z} + \frac{1}{z-1}\right) u' + \left(\frac{\mu}{z} + \frac{\nu}{z-1}\right) u = 0,$$

with

$$z_0 = \texttt{0.5 + 0.150755672288881811323406 i}, \qquad \alpha = \frac{2i\sqrt{11}}{3\sqrt{3}},$$

$$\mu = \frac{-5 - i\sqrt{11}}{54} + \frac{i\sqrt{11}}{3\sqrt{3}}, \qquad \nu = \frac{5 - i\sqrt{11}}{54} + \frac{i\sqrt{11}}{3\sqrt{3}},$$

where the CM motivation is the discriminant-11 η-geometry at $\tau = i\sqrt{11}/3$.

The numerical bottleneck in this search is the evaluation of the regular local Heun branches at z_0. An earlier RK4 path transport gave only `7.642` digits of branch stability, so that computation was useful only as a heuristic probe. We therefore replaced it by local regular-series recurrences about $z = 0$ and $z = 1$, matched at z_0; at `120` decimal digits and continued-fraction depth `9000`, the two independent local evaluations now agree to `35.207` digits. Since this stability margin exceeds the best observed near-miss by almost 30 digits, the null PSLQ output is no longer plausibly attributable to solver drift.

We then searched for low-height relations

$$aV_{\text{quad}} + bH + cS + dC + e = 0,$$

where H ranges over the regular confluent-Heun invariants at z_0, S over $\{y(0), y'(0), y'(0)/y(0), 1/(y'(0)/y(0)), M_{11}, M_{21}, M_{11}/M_{21}, M_{12}/M_{22}\}$, and C over the CM basis $\{\pi/\sqrt{11}, \pi^2/11, \Gamma(1/11)\Gamma(2/11)\Gamma(4/11)/\Gamma(8/11), \sqrt{11}, \log|\eta(\tau)|\}$. A total of `360` triples were tested with PSLQ coefficient bound `2000`; the number of non-trivial hits involving V_{quad} was `0`. The strongest low-height near-miss was

$$V_{\text{quad}} \approx -\Im\left(\frac{H_0'(z_0)}{H_0(z_0)}\right) + \frac{3}{2}\,y(0) + 3\log|\eta(\tau)| - 1, \tag{1}$$

with accuracy only `5.507602` digits.

For the exclusion table: the hardened three-basis confluent-Heun/Stokes/η search at `120` dps gives `360` tested triples, `0` non-trivial PSLQ hits, and best near-miss `5.507602` digits.

Modular input for Lemma K. The Kloosterman-type sum $A_k(1,q) = \sum_{d \bmod q,\, \gcd(d,q)=1} \exp(-2\pi i k\,\theta(d,q))\, e^{2\pi i d^{-1}/q}$, where $\theta(d,q) = (a+d)/(12q) - s(d,q)$ and $a \equiv d^{-1} \pmod q$, is attached to the $\eta(\tau)^{-k}$ multiplier via Apostol [**?**] Theorem 3.4 (see also Iwaniec–Kowalski [**?**] §2.8); the sign $\exp(-2\pi i k\,\theta)$ is the correct inverse character. Since the eta multiplier has unit modulus, Weil's bound for classical Kloosterman sums gives $|A_k(1,q)| \ll d(q)q^{1/2}$ uniformly in k. In our explicit sweep for $1 \le k \le 24$ and $q \le 1000$ we find

$$C_k := \max_{q \le 1000} \frac{|A_k(1,q)|}{d(q)q^{1/2}} \le 1.4368,$$

and the values are exactly 12-periodic in k, consistent with the underlying 24th-root eta multiplier.

8 Further Structure of the Convergent Numerators

Define $R(n,m) = p_n(m)/(2n-1)!!$ as in Conjecture 1. For $m = 0$ the closed form is $R(n,0) = 2n+1$, equivalently $2\,\Gamma(n+\frac{3}{2})/\Gamma(n+\frac{1}{2})$—elementary, but the continuous extension to all real m (Theorem **??**) is not. For $m = 1$, $R(n,1) = n^2+3n+1$ (OEIS A028387 [**?**], discriminant 5), whose reciprocal sum satisfies the identity

$$\sum_{n=1}^{\infty} \frac{1}{n^2+3n+1} = 1 + \frac{\pi \tan(\sqrt{5}\,\pi/2)}{\sqrt{5}}.$$

For $m \geq 2$, $R(n,m)$ is rational with denominators dividing the product of odd primes $\leq 2n-1$. As a function of m for fixed n, $R(n,m)$ is polynomial of degree $\lfloor n/2 \rfloor$ with coefficients $c_k(n)$ satisfying $c_0(n) = 2n+1$ universally and, for $k \geq 1$,

$$\operatorname{den}(c_k(n)) \;\Big|\; \prod_{\substack{p \leq 2n-1 \\ p \text{ odd prime}}} p^{a_p(n)}.$$

This structure is consistent with a hypergeometric representation of $R(n,m)$ involving half-integer parameters, but the exact closed form remains open.

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